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An Efficient DDoS Attack Detection in SDN using Multi-Feature Selection and Ensemble Learning

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Abstract

Software-defined networking (SDN) has significantly enhanced network agility by decoupling network control from hardware devices. This architectural shift exposes SDN infrastructures to increased risks from Distributed Denial of Service (DDoS) attacks, which can severely disrupt network services and render them unavailable to legitimate users. Despite the existence of various techniques for detecting specific types of DDoS attacks, there remains a need for more comprehensive work capable of accurately identifying multiple attack categories. We have addressed the gap by proposing a robust DDoS attack detection method in the SDN. We utilize advanced machine learning (ML) algorithms to analyze the InSDN dataset, employing advanced feature extraction techniques to improve detection accuracy and efficiency. Three feature extraction methods, SelectKBest, ANOVA F-value scores, and feature importance scores from RandomForest Classifier, were used to select the best ten features from 81. This feature selection, combined with ensemble learning, achieved an accuracy of 99.9%. The Receiver Operating Characteristic (ROC) curve, confusion matrix, and K-fold cross-validation were used to evaluate the performance of the proposed model.

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1. Introduction

SDN transforms network management by separating the control plane from the data plane, enabling centralized control and network automation. However, this centralization introduces significant security challenges. The SDN controller becomes a critical attack target; if compromised, an attacker can manipulate the entire network. Additionally, the communication channels between the controller and network devices, often reliant on standard protocols like OpenFlow, can be susceptible to interception and tampering. Furthermore, the abstraction layers in SDN can obscure visibility, complicating traditional security measures such as monitoring and intrusion detection. These vulnerabilities necessitate robust security strategies, including encryption, authentication, and continuous monitoring, to protect SDN environments effectively [1].

DDoS attacks are popular cyber attacks on SDN networks as they are particularly vulnerable due to their centralized architecture [2]. Attackers can overwhelm the SDN controller with excessive malicious traffic, disrupting its ability to effectively control the network and compromising overall system performance. To mitigate these threats, SDN can use its programmability to implement dynamic and automated responses. Anomaly detection

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algorithms can be incorporated into the controller to recognize abnormal traffic patterns that suggest a DDoS attack. Once detected, the controller can dynamically reconfigure the network to isolate and mitigate the impact of the attack, such as rerouting traffic, rate limiting, or deploying honeypots. Additionally, SDN's centralized nature allows for more efficient coordination of defense mechanisms across the entire network, enhancing its resilience against large-scale DDoS attacks[3].

Most of the literature on DDoS attack detection in SDN has focussed on ML and Deep Learning (DL) techniques with little focus on feature selection[4]. Most existing studies focus on singular detection techniques, often neglecting the potential of multi-feature selection and ensemble learning methods, which can offer enhanced accuracy and resilience against diverse attack vectors. Furthermore, there is less focus by researchers that thoroughly examine the integration of these advanced methodologies within the SDN framework, addressing the dynamic and adaptive nature of modern cyber threats [5]. Consequently, a comprehensive approach that integrates multi-feature selection but the ensemble learning remains unexplored.

This research study aims to utilize advanced ML techniques, particularly multi-feature selection and ensemble learning, to improve DDoS attack detection in SDN environments. The multi-feature selection process incorporates SelectKBest, ANOVA F-value scores, and feature importance scores from the RandomForest Classifier, which notably enhances the model's accuracy and efficiency. Furthermore, the core contribution of this work lies in the integration of multi-feature selection with ensemble learning to form a comprehensive and efficient DDoS attack detection system for SDN environments. Our proposed model outperforms traditional single-algorithm approaches by combining the strengths of multiple learning algorithms, thus enhancing detection accuracy and reducing false positives. Additionally, the use of ensemble methods ensures adaptability to evolving attack patterns, providing a dynamic and resilient security solution. Through extensive simulations, we demonstrate the performance of our approach. The primary contributions of this paper are outlined as follows.

- Carried out comprehensive analysis of the InSDN dataset, incorporating various features and pre-processing techniques.
- Introduced feature selection mechanism that identifies the top 10 features out of 81 using multiple feature selection approaches.
- Ensemble learning was employed to detect DDoS attacks in SDN, and its performance was compared against other ML algorithms for classifying network traffic as either malicious or benign.

The structure of the paper is organized as follows: Section 2 reviews the existing literature on DDoS attack detection mechanisms in SDN. Section 3 describes the proposed system for detecting DDoS attacks using a multi-feature selection method and an ensemble learning classifier. Section 4 presents the implementation results and evaluates the system's performance, highlighting metrics such as the ROC curve and K-fold cross-validation. Finally, Section 5 provides the conclusion of the study.

2. Related Works

The authors in [6], have proposed a flow-based anomaly detection method for identifying DDoS attacks in SDN, using feature selection techniques. The approach applies Random Forest (RF) and Information Gain (IG) to extract key features from the CICIDS2018, CICIDS2017, and InSDN datasets with accuracy of 99.1% for LSTM model. In [7], authors have proposed an ensemble of K-means clustering and RF algorithm that involves clustering network traffic data using K-means to identify potential anomalies and then applying the RF classifier to accurately detect DDoS attacks. The dataset used for the experiments includes both synthetic and real-world SDN traffic data. The model was evaluated on the CICIDS2017 and obtained an accuracy of 99.12%.

The authors of [8], have proposed a majority voting average for feature selection combined with various ML approaches using the InSDN dataset, containing various DDoS attack types. The study evaluated seven machine learning classifiers, including Support Vector Machine (SVM), XGBoost, Multi-Layer Perceptron (MLP), Decision Tree (DT), Naïve Bayes (NB), RF, and K-Nearest Neighbor (KNN). The RF classifier achieved the highest performance of 99.85% accuracy and a 0.002 False Positive Rate. The study employs a comprehensive approach to detect DDoS attacks within SDN environments using ensemble-based ML techniques. In [9], authors have proposed a methodology that employs the RF algorithm with hyperparameter tuning and recursive feature elimination to select the most significant features from the dataset. The dataset used for this research includes 207,146 traffic flows with 84 features, capturing a variety of normal and attack traffic types such as DoS, DDoS, Probe, BFA, Web-Attack, and BOTNET with model's performance for RF of 99.85% accuracy.

In article [10], authors present a classifier system designed to detect and prevent DDoS TCP flood attacks in cloud environments. This system employs four classifiers—least squares support vector machine (LS-SVM), KNN, NB, and MLP to classify incoming packets and identify malicious traffic, achieving an accuracy of 96.8%.

In paper [11], authors have proposed a combined SVM-KNN algorithm for detecting DDoS attacks in SDN. The methodology selects five key features from packet headers and utilizes the SVM-KNN algorithm for training and classification. The dataset, generated using the Mininet emulator with the Ryu controller, simulates both normal and attack traffic, achieving an accuracy of 97.36%. The authors of [12], have proposed two different modules: one for low-rate Slow-TCAM attacks using the Alleviative Threat for TCAM (ATFT) algorithm, and another for high-rate DDoS attacks using ensemble learning techniques with 99% accuracy rate and framework was tested in a Mininet simulation environment, and assessed using the CICDDoS2019 dataset.

The authors of [13], have proposed anomaly behavior detection with principle component analysis involves calculating the entropy of network traffic to identify anomalies and generate a dataset for ML models. Features such as packet count and byte flow were extracted from the network. Various classifiers, including SVM and DT, were evaluated, with SVM achieving an accuracy of 97%. In article [14], authors have proposed an efficient DDoS attack detection mechanism in SDN environments using Deep Convolutional Neural Networks (DCNN). The model is trained and evaluated using three datasets: InSDN, CIC-IDS2017, and CIC-DDoS2019. The proposed DCNN model achieved accuracy, with detection rates of 99.9% for InSDN, 99.8% for CIC-IDS2017, and 99.7% for CIC-DDoS2019.

The authors of [15], have used In-band Network Telemetry (INT) for switch state sensing in the Spatial-Temporal Graph Convolutional Network (ST-GCN), which maps the network into a graph. The dataset used includes data collected through INT with sampling, capturing various network state indicators. The detection model processes both spatial and temporal features of network state changes to identify DDoS attack paths. In article [16], authors have proposed combining entropy-based detection with ML to identify DDoS attacks in SDN. In order to detect possible DDoS attacks early on, the suggested two-level technique first analyzes network traffic unpredictability using the entropy method. If an anomaly is detected, ML models such as NB, SVM, and RF are employed for further verification. With an accuracy of 98.3% for RF in both training and evaluation, the study shows the efficacy of the InSDN dataset in DDoS detection while preserving a low false positive rate. The dataset consists of both normal and attack traffic.

The authors of [17], have proposed a novel approach for detecting Advanced Persistent Threat (APT) scanning activities in SDN. The methodology involves enhancing the SDNmap tool to emulate stealthier APT scanning techniques, followed by the application of ML algorithms for detection. The dataset used includes traffic generated in Mininet, reflecting various network sizes and scenarios. In paper[18], the study employs the XGBoost classifier, achieving high-performance metrics with at least 97.8% accuracy using five features. The authors have employed the feature selection mechanism using the Extra Trees Classifier, and the selected features are fed into DT, XGBoost, and RF classifiers. The XGBoost classifier achieved an accuracy of 98.72% for CICDDoS2019 dataset.

The authors of [19], have proposed a method for detecting anomaly flows in SDN using the DPTCM-KNN algorithm. This method improves the existing TCM-KNN algorithm by incorporating both strangeness and independence to enhance anomaly detection accuracy. The dataset consists of network traffic data collected in a simulated SDN environment using Mininet and the Ryu controller, capturing both normal and attack traffic with an accuracy of 98.2%. In the paper [20], the hybrid mechanism of K-means++ and Fast K-Nearest Neighbors (K-FKNN) methodology that involves first clustering training data with K-means++ to obtain centroids and radii for each cluster, followed by using FKNN for flow detection to improve efficiency and accuracy with NSL-KDD dataset. The experimental results demonstrate that K-FKNN significantly enhances detection speed and accuracy compared to traditional KNN, achieving high precision and stability in DDoS detection.

The authors of [21], present a Factorization Machine (FM) approach for identifying low-rate DDoS attacks in SDN. To detect low-rate DDoS attacks, this method takes characteristics from flow rules, such as packet numbers, duration time, packet interval, and relative dispersion of match bytes. The collection comprises traffic data produced in a Mininet and Ryu-simulated SDN system. The FM-based detection approach provides a detection accuracy of 95.80%, according to experimental results. In article [22], the authors have proposed a methodology that involves constructing an SDN environment using Mininet and Floodlight, extracting six-tuple characteristic values from switch flow tables. These features include the speed of the source IP and port, standard deviation of flow packets and bytes, speed of flow entries, and the ratio of pair-flow. The dataset, which consists of normal and attack traffic generated using the Hping3 tool, was analyzed using an SVM-based detection model, achieving an average accuracy rate of 95.24%.

The authors of [23], introduced an ML framework that is used in the defense system to mitigate DDoS attacks in SDN-based networks. To identify novel flow attacks and common DDoS attacks, the system uses ML classification algorithms to gather network statistics from forwarding elements on a regular basis. Training and testing of the system with 99.4% classification accuracy is conducted using the NSL-KDD dataset. In article[24], introduced the Renyi entropy technique, network traffic is tracked at the SDN controller. The Mininet emulator and Ryu controller were used to create the dataset, which simulated both legitimate and malicious traffic.

Another framework, as proposed in [25] unique flow table sharing technique is used as a defense measure to mitigate DDoS attacks in SDN-based cloud systems. In order to minimize overload during DDoS attacks, this methodology shares unused flow table space across switches and represents flow table space using a mathematical model based on queuing theory. The dataset used for evaluation includes simulation-based experiments conducted in a Mininet environment. The results demonstrate that the proposed approach significantly enhances the resistance of SDN against flow table overloading DDoS attacks, reducing the number of OpenFlow messages exchanged and increasing the holding time of SDN during attacks. In article [26], authors have proposed a meta-heuristics approach called LION IDS for detecting DDoS attacks in SDN using the Lion Optimization Algorithm (LOA). The methodology involves applying LOA for feature selection and Convolutional Neural Networks (CNN) for classification. The dataset used for evaluation is the NSL-KDD dataset, formatted to reflect real-time traffic from a Mininet environment with a detection accuracy of 98.2%.

3. Proposed Method

Our proposed method in Fig. 1 provides a thorough analysis of the InSDN dataset, focusing on a wide range of features and employing advanced pre-processing techniques. The dataset is analysed to identify key characteristics relevant to detecting DDoS attacks in SDN environments. Pre-processing techniques, including feature selection, noise reduction, and data normalization, are used to improve the accuracy and dependability of the data. This comprehensive approach ensures that the most pertinent features are retained, facilitating the development of robust and accurate detection models. The analysis aims to optimize the dataset for ML algorithms, improving their performance in identifying and mitigating DDoS attacks.

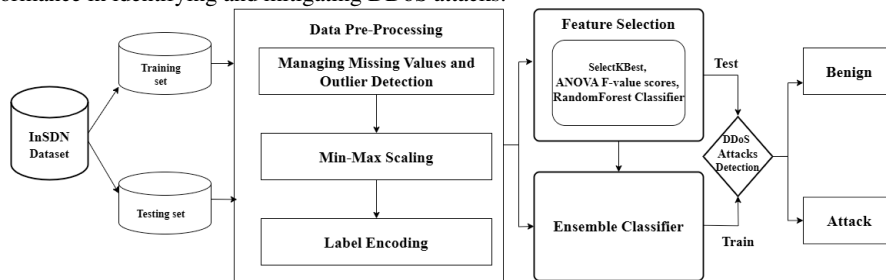


Fig. 1: Proposed system model for Ensemble Classifier

The authors of [27], have proposed a comprehensive methodology for generating an SDN-specific dataset called InSDN. The authors created a virtualized network testbed using Mininet, an SDN controller (ONOS), and various virtual machines to simulate real network environments and generate diverse traffic data, including normal and attack scenarios. The dataset encompasses multiple attack types, such as DDoS, web attacks, botnets, and exploitation, generated using tools like Kali Linux and Metasploitable2. The features extracted from the traffic data include flow statistics like duration time, packet count, and byte count, which were standardized and encoded for machine learning model training. The InSDN dataset's efficacy was demonstrated through the application of various ML techniques, achieving high accuracy rates in detecting anomalies in SDN environments.

3.1. Data Pre-processing

The study introduces a novel feature selection mechanism aimed at identifying the top 10 features from an initial set of 81. This mechanism combines three feature extraction methods: SelectKBest, ANOVA F-value scores, and feature importance scores from a RandomForestClassifier. By integrating these methods, the approach effectively identifies the most relevant features, improving both the performance and accuracy of the detection model. This comprehensive feature selection process ensures that only the most critical attributes are retained, optimizing the model's ability to detect anomalies in SDN environments. SelectKBest, in particular, ranks features based on their importance using statistical testing, such as ANOVA F-value, allowing the model to retain the most informative features while reducing the dataset's dimensionality.

ANOVA F-value scores evaluate the significance of features by comparing the variance between groups to the variance within groups, providing an F-score for each feature. Features with higher F-scores have a greater influence on the target variable, aiding in selecting the most relevant features. Similarly, a RandomForestClassifier ranks feature importance based on the reduction in impurity each feature brings across the decision trees. Features with higher importance scores contribute more to model accuracy. By combining these methods, the model effectively identifies and selects the most critical features for accurate DDoS attack detection in SDN environments.

Let $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$ be the sets of features selected by SelectKBest, ANOVA F-value scores, and RandomForest-Classifer, respectively, each containing 20 features:

$$\mathcal{S}_1 = \{f_{1,1}, f_{1,2}, \dots, f_{1,20}\} \quad (1)$$

$$\mathcal{S}_2 = \{f_{2,1}, f_{2,2}, \dots, f_{2,20}\} \quad (2)$$

$$\mathcal{S}_3 = \{f_{3,1}, f_{3,2}, \dots, f_{3,20}\} \quad (3)$$

To find the top 10 frequently occurring features among these sets, we follow these steps:

1. **Feature Frequency Count:** Calculate the frequency of each feature across the three sets. Let $\mathcal{F}(f_i)$ represent the frequency of feature f_i :

$$\mathcal{F}(f_i) = \sum_{k=1}^3 \mathbb{I}(f_i \in \mathcal{S}_k) \quad (4)$$

where $\mathbb{I}(\cdot)$ is the indicator function, which is set to 1 in the case of a true condition and 0 in the absence of one.

2. **Rank Features by Frequency:** Rank all features f_i based on their frequency $\mathcal{F}(f_i)$ in descending order. Let $\mathcal{R}$ be the ordered set of features based on their frequency:

$$\mathcal{R} = \{f_{r_1}, f_{r_2}, \dots, f_{r_{81}}\} \quad \text{such that} \quad \mathcal{F}(f_{r_1}) \geq \mathcal{F}(f_{r_2}) \geq \dots \geq \mathcal{F}(f_{r_{81}}) \quad (5)$$

3. **Select Top 10 Features:** Choose the top 10 features with the highest frequency from the ranked set $\mathcal{R}$:

$$\mathcal{T} = \{f_{r_1}, f_{r_2}, \dots, f_{r_{10}}\} \quad (6)$$

features are consistently identified as important.

The set $\mathcal{T}$ represents the top 10 frequently occurring features selected from the three methods, ensuring that these

Table 1: Top 10 features selected from the InSDN dataset

Sl. No	Name of Feature	Description of Feature
1	Src IP	IP address from which the packet is originating
2	Dst IP	IP address to which the packet is destined
3	Protocol	Communication protocol used (e.g., TCP, UDP)
4	Pkt Len Std	Standard deviation of packet lengths in the flow
5	Flow ID	Unique identifier assigned to each flow
6	Bwd Pkt Len Mean	Average length of packets in the reverse direction
7	Pkt Len Var	Variance in the packet lengths within the flow
8	Src Port	Source port number used by the sender
9	Bwd Seg Avg Size	Average segment size in the reverse direction
10	Bwd Pkt Std Len	Standard deviation of packet lengths in the reverse direction

In this study, we employ several classification methods to identify DDoS attacks within the InSDN dataset, focusing on the top 10 selected features. XGBoost (Extreme Gradient Boosting) is used as an ensemble learning technique, which integrates the predictions of multiple weak models to form a robust predictive model. The XGBoost classifier constructs an additive model incrementally, optimizing the objective function iteratively.

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a dataset with n samples, where x_i represents the feature vector and y_i represents the corresponding target value.

1. Objective Function:

The objective function to be minimized in XGBoost consists of a loss function L and a regularization term Ω :

$$\mathcal{L}(t) = \sum_{i=1}^n L(y_i, \hat{y}_i^{(t)}) + \sum_{k=1}^t \Omega(f_k) \quad (7)$$

where $\hat{y}_i^{(t)}$ is the prediction for the i -th instance at the t -th iteration, and f_k represents the k -th tree.

2. Additive Model:

The model is built in an additive manner. At each iteration t , a new function f_t (decision tree) is added to minimize the objective function:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i) \quad (8)$$

3. Loss Function:

The difference between the actual value y_i and the expected value $\hat{y}_i$ is measured by the loss function. A common choice for the loss function in classification tasks is the logistic loss:

$$L(y_i, \hat{y}_i) = -[y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \quad (9)$$

4. Regularization Term:

The term regularization Ω regulates the model's complexity to avoid overfitting. For tree models, the regularization term can be expressed as:

$$\Omega(f_t) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (10)$$

5. Gradient and Hessian:

Second-order Taylor expansion is used by XGBoost to optimize the objective function. The gradients (first-order derivative) $g_i^{(t)}$ and Hessians (second-order derivative) $h_i^{(t)}$ are computed as follows:

$$g_i^{(t)} = \frac{\partial L(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}} \quad (11)$$

$$h_i^{(t)} = \frac{\partial^2 L(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)2}} \quad (12)$$

6. Tree Structure:

The structure of the tree is determined by maximizing the gain, which is computed based on the gradients and Hessians:

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \quad (13)$$

where G_L and G_R are the sums of gradients and H_L and H_R are the sums of Hessians for the left and right nodes, respectively.

By iteratively adding trees and optimizing the objective function, the Ensemble classifier effectively models complex relationships in the data, providing robust and accurate predictions for DDoS attack detection in the InSDN dataset.

3.2. DDoS Attacks detection using Ensemble Classifier

The proposed algorithm evaluates an ensemble classifier using K-Fold cross-validation, ensuring reliable performance metrics. It begins with initializing performance measurements and capturing current time, CPU, and memory usage. The dataset D is loaded into a DataFrame and split into features X and labels y , then further divided into training and testing sets using an 80-20 split. The algorithm defines and trains models by initializing the HistGradientBoostingClassifier and XGBClassifier, which are combined into a VotingClassifier for soft voting. Averaging the results from each fold provides a reliable estimate of the model's accuracy and generalization capability. The algorithm performs K-Fold cross-validation with five splits, shuffling the data to

ensure robust evaluation. It calculates and reports cross-validation results, including mean accuracy and standard deviation. The model is then evaluated on the test set to determine accuracy and generate a classification report.

Algorithm 1 Ensemble Classifier for DDoS Attack Detection and Performance Evaluation

```

1: Input: Dataset  $D$ , Features  $X$ , Labels  $y$ 
2: Output: Model Performance Metrics, Saved Model
3: procedure INITIALIZE PERFORMANCE MEASUREMENT
4:    $t_0 \leftarrow \text{current\_time}()$ 
5:    $CPU_{before} \leftarrow \text{measure\_cpu}()$ 
6:    $MEM_{before} \leftarrow \text{measure\_memory}()$ 
7: end procedure
8: procedure LOAD AND SPLIT DATA
9:   Load dataset  $D$  into DataFrame
10:   $X \leftarrow D.\text{drop}(\text{'Label'})$ 
11:   $y \leftarrow D[\text{'Label'}]$ 
12:   $(X_{train}, X_{test}, y_{train}, y_{test}) \leftarrow \text{train\_test\_split}(X, y, \text{test\_size} = 0.2, \text{random\_state} = 42)$ 
13: end procedure
14: procedure DEFINE AND TRAIN MODELS
15:   Initialize  $hgb\_clf \leftarrow \text{HistGradientBoostingClassifier}()$ 
16:   Initialize  $xgb\_clf \leftarrow \text{XGBClassifier}()$ 
17:    $\text{voting\_clf} \leftarrow \text{VotingClassifier}(\text{estimators}=[('hgb', hgb\_clf), ('xgb', xgb\_clf)], \text{voting}='soft')$ 
18:    $\text{voting\_clf.fit}(X_{train}, y_{train})$ 
19: end procedure
20: procedure PERFORM K-FOLD CROSS-VALIDATION
21:    $kfold \leftarrow \text{KFold}(n\_splits = 5, \text{shuffle} = \text{True}, \text{random\_state} = 42)$ 
22:    $cv\_results \leftarrow \text{cross\_val\_score}(\text{voting\_clf}, X, y, cv = kfold, \text{scoring} = \text{'accuracy'})$ 
23:   Print  $cv\_results, cv\_results.mean(), cv\_results.std()$ 
24: end procedure
25: procedure EVALUATE MODEL ON TEST SET
26:    $y_{pred} \leftarrow \text{voting\_clf.predict}(X_{test})$ 
27:    $\text{accuracy} \leftarrow \text{accuracy\_score}(y_{test}, y_{pred})$ 
28:    $\text{report} \leftarrow \text{classification\_report}(y_{test}, y_{pred})$ 
29:   Print  $\text{accuracy}, \text{report}$ 
30: end procedure
31: procedure SAVE MODEL AND MEASURE PERFORMANCE
32:   Save model:  $\text{joblib.dump}(\text{voting\_clf}, \text{'ensemble\_model\_latest.pkl'})$ 
33:    $CPU_{after} \leftarrow \text{measure\_cpu}()$ 
34:    $MEM_{after} \leftarrow \text{measure\_memory}()$ 
35:    $t_1 \leftarrow \text{current\_time}()$ 
36:    $\text{execution\_time} \leftarrow t_1 - t_0$ 
37:   Print  $\text{execution\_time}, CPU_{before}, CPU_{after}, MEM_{before}, MEM_{after}$ 
38: end procedure

```

Overall, this approach provides a comprehensive evaluation of the model's performance and consistency, ensuring that the ensemble model is both accurate and reliable for detecting network anomalies in SDN environments. These findings highlight the effectiveness of combining traditional and advanced ML techniques for accurate DDoS attacks detection in SDN environments. Selecting the top 10 features from the InSDN dataset significantly enhances model performance, ensuring robust and precise classification.

To ensure robust model evaluation, the code implements cross-validation of 5-fold using the 'cross_val_score' function from 'sklearn.model.selection'. This technique divides the dataset into five subsets, trains the model on four subsets, and validates it on the remaining subset. This process is repeated five times, each time with a different validation set, and the results are averaged to provide a more reliable estimate of model performance. The mean cross-validation accuracy achieved by the ensemble learning is 99.8%, demonstrating its high efficacy in detecting DDoS attacks across different dataset partitions.

The evaluation of an ensemble ML model to detect DDoS attacks in SDN. The key steps involve measuring system resources before and after execution, splitting the dataset into training and testing sets, training the ensemble model, evaluating its performance, and saving the trained model. The results indicate that the ensemble model was successfully trained and saved as 'ensemble_model_latest.pkl'. The entire process took approximately 11.09

seconds to execute, with CPU utilization increasing from 1.9% to 46.8% during model training and evaluation, reflecting the computational demand of these tasks.

4. Results and Discussion

The model's performance is assessed using metrics like accuracy, precision, recall, and F1-score that are derived from a classification report on the predictions test set as represented in the above graphs [28]. Key performance indicators include execution time, CPU utilization, and memory usage before and after the process, providing a comprehensive view of the model's computational efficiency. The ensemble model's efficacy in identifying DDoS attacks is demonstrated by its thorough performance metrics and overall accuracy. The trained model is saved using joblib for future deployment.

4.1. Experimental Setup

Table 2 provides an overview of the hardware specifications of the Lenovo ThinkPad P53 Workstation used in the implementation. The system features 12 cores, ensuring robust parallel processing capabilities. The workstation is equipped with 31.75 GB of RAM, out of which 21.02 GB was utilized. This high-performance configuration supports the computational demands of running complex ML algorithms and processing large datasets effectively.

Table 2: System Hardware Specifications

Specification	Details
System Details	Lenovo ThinkPad P53 Workstation
No. of Cores	12
Total RAM	32 GB

4.2. Performance Analysis

The performance analysis subsection assesses the efficiency of the proposed ensemble model for DDoS attack detection in SDN by applying different feature selection techniques to enhance accuracy and model performance. It highlights how combining multiple feature selection techniques significantly enhances the model's accuracy.

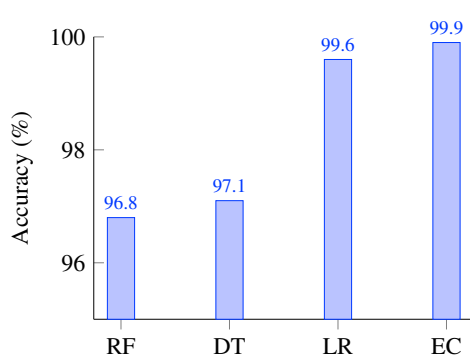


Fig. 2: Accuracy Comparison of Classification Algorithms

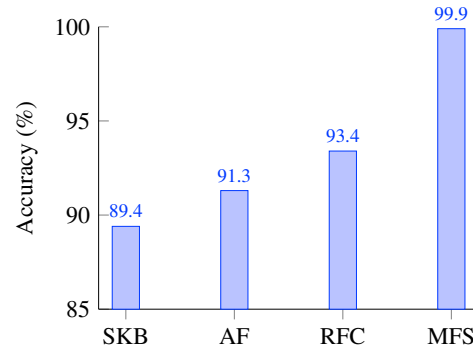


Fig. 3: Accuracy of Ensemble Classifier using different feature selection methods

In order to detect DDoS attacks in SDN, this methodology evaluates an ensemble ML model employing specific features. The Fig.2. illustrates the accuracy percentages of the Ensemble classifier using different feature selection methods. The Ensemble classifier utilizing the SelectKBest feature selection achieved an accuracy of 89.4%, which was the lowest among the compared methods. When the ANOVA F-Value scores were used for feature selection, the classifier's accuracy slightly improved to 91.3%. The Random Forest feature importance method, which resulted in an accuracy of 93.4%. The highest accuracy, 99.9%, was achieved using a multi-feature selection approach that likely integrates multiple feature selection techniques. The fig.3 illustrates the results of Ensemble classifier with combination of different feature selection techniques like SelectKBest (SKB), ANOVA F-value scores(AF), feature importance scores from RandomForest Classifier (RFC), and Multi-Feature Selection (MFS). The ensemble model integrates a Histogram-Based Gradient Boosting (HGBClassifier) and an XGBoost classifier (XGBClassifier) within a VotingClassifier framework, considering their strengths to enhance detection accuracy and robustness.

The dataset, containing pre-selected features for DDoS attack detection, is divided into training and testing sets using an 80-20 split, ensuring a comprehensive and reliable evaluation of the model's performance. The HGB-Classifier and XGBoost classifiers are then trained on the training data, with the VotingClassifier combining their predictions using a soft voting strategy. The graph illustrates the accuracy percentages of the Ensemble classifier using different feature selection methods. The Ensemble classifier utilizing the SelectKBest feature selection method achieved an accuracy of 89.4%, which was the lowest among the compared methods. When the ANOVA F-Value scores were used for feature selection, the classifier's accuracy slightly improved to 91.3%. A more significant improvement was observed with the RF feature importance method, which resulted in an accuracy of 93.4%. The highest accuracy, 99.9%, was achieved using a multi-feature selection approach that likely integrates multiple feature selection techniques using Ensemble classifier(EC). This approach substantially enhanced the classifier's performance, highlighting the effectiveness of combining various feature selection methods to improve detection accuracy. Overall, the graph demonstrates that employing a combination of feature selection techniques yields the best performance for the Ensemble classifier in terms of accuracy.

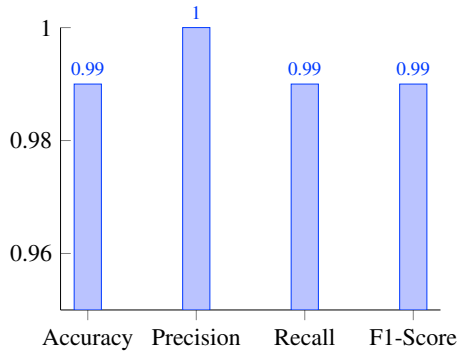


Fig. 4: Performance evaluation of Ensemble Classifier

True Label	0	1
0	13,619	0
1	1	27,414
	0	1

Fig. 5: Confusion Matrix of Ensemble Classifier

The methods include DT, Logistic Regression (LR), RF, and an ensemble classifier approach. LR achieves an accuracy of 99.6%, providing a probabilistic interpretation of the results. DT, known for its clear and visual decision-making process, achieves an accuracy of 97.1%. RF, which mitigates overfitting by averaging multiple decision trees, achieves an accuracy of 96.8%. The Ensemble classifier, recognized for its efficiency and advanced regularization, delivers the highest accuracy at 99.9%. These results highlight the superior performance of the Ensemble classifier in accurately classifying network traffic and detecting DDoS attacks, making it a robust choice for such tasks. Additionally, the high accuracy of LR demonstrates its effectiveness, while DT and RF also provide reliable results with their respective advantages in interpretability and robustness. The confusion matrix in fig.5 for the Voting Classifier highlights its exceptional performance in classifying network traffic with 13,619 instances correctly identified as benign traffic, 0 instances incorrectly identified as malicious, 1 instance incorrectly identified as benign, and 27,414 instances correctly identified as malicious traffic.

This approach substantially enhanced the classifier's performance, highlighting the effectiveness of combining various feature selection methods to improve detection accuracy. Overall, the graph demonstrates that employing a combination of feature selection techniques yields the best performance for the Ensemble classifier in terms of accuracy.

5. Conclusion and Future Scope

Our proposed methodology highlights the effectiveness of applying machine learning algorithms to detect DDoS attacks in SDN systems using the InSDN dataset. The proposed model integrates several feature selection methods, including SelectKBest, ANOVA F-value scores, and feature importance scores from the RandomForest-Classifier, to identify the top 10 most relevant features from a total of 81. The Ensemble classifier achieved the highest accuracy of 99.9%, surpassing other machine learning models such as DT, RF and LR. The robustness of the proposed model was evaluated across different DDoS attack scenarios, confirming its versatility and effectiveness in real-world applications. This research highlights the importance of using specialized datasets like InSDN to enhance SDN security, providing a comprehensive approach to anomaly detection that can be pivotal for maintaining network integrity. The future work will concentrate on the real-time implementation of the proposed DDoS detection model to handle large-scale SDN deployments. Further optimization of the model will be pursued to enhance its efficiency and scalability. A crucial area of research will be the creation of adaptive algorithms that can dynamically react to changing DDoS attack patterns for the ongoing security of SDN systems.

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